

Assignment - 4

(1) Vector $A = \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$

(a) The dimensions of the subspaces according to the fundamental theorem are:-

① Column space $(C(A)) = 1$

② Row space $(C(A^T)) = 1$

③ Null space $(N(A)) = \text{N/A}$ 0

④ Left null space $(N(A^T)) = 2$

(b) Bases for the 4 fundamental subspaces

01 ~~Column space~~ $(C(A))$

① Column space $(C(A))$:- The vector A itself
~~basis of~~

Basis for column space = $\begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$

② Null space ($N(A)$) :- It has no non-zero vectors.
Basis of null space = $\{0\}$

③ Row space ($C(A)$) :- It is same as A^T

$$\text{Basis of row space} = \left\{ (-2 \ 4 \ -2) \right\}$$

④ Left null space ($N(A^T)$) :-

for this we need to solve for $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ such that

$$(-2 \ 4 \ -2) \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 0$$

$$\therefore -2v_1 + 4v_2 - 2v_3 = 0$$

$$\therefore v_1 - 2v_2 + v_3 = 0$$

$$\therefore v_1 = 2v_2 - v_3$$

$$\therefore v = \begin{pmatrix} 2v_2 - v_3 \\ v_2 \\ v_3 \end{pmatrix}$$

if $v_2 = 1$ and $v_3 = 0$, if $v_2 = 0$, $v_3 = 1$

$$v = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$$v = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\therefore \text{Basis of left null space} = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

(c) cross product of the null space of A^T basis vectors and compare the results to column space of A

$$v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$v_1 \times v_2 = \begin{vmatrix} i & j & k \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = i(1-0) - j(2-0) + k(0+1)$$

$$= i - 2j + k$$

$$\therefore v_1 \times v_2 = \begin{pmatrix} +1 \\ -2 \\ 1 \end{pmatrix}$$

Comparing it with A , the cross product vector is a scalar multiplier of A

Q2]

$$A_{2 \times 3} = \begin{pmatrix} -2 & 4 & -2 \\ -2 & 4 & -2 \end{pmatrix}$$

(a) Dimensions of subspaces are :-

(1) column space $(C(A)) = \text{rank of } A$
 $= 1$

(2) Null space $= n - \text{rank}(A) = 3 - 1 = 2$

(3) row space $= d(C(A^T)) = \text{rank}(A) = 1$

(4) left null space $= m - \text{rank}(A) = 2 - 1 = 1$

(b) Obtain bases for the 4 fundamental subspaces.

① Basis of column space $\Rightarrow$ The column vector space is spanned by non-zero columns of A . Since both columns are identical

$$\text{column space basis} = \left\{ \begin{pmatrix} -2 \\ -2 \end{pmatrix} \right\}$$

② ~~Null~~ Null space ($N(A)$)

The null space consists of vectors $x \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ such that

$$Ax = 0$$

$$\therefore \begin{pmatrix} -2 & 4 & -2 \\ -2 & 4 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\therefore -2x_1 + 4x_2 - 2x_3 = 0$$

$$\therefore x_1 - 2x_2 + x_3 = 0$$

$$\therefore x_1 = 2x_2 - x_3$$

$$\therefore x = \begin{pmatrix} 2x_2 - x_3 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\therefore \text{If } x_2 = 0 \text{ \& } x_3 = 1$$

$$x = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{If } x_2 = 1 \text{ \& } x_3 = 0$$

$$x = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$$\therefore \text{Null space basis} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

(3) Left null space $(N(A^T))^{\perp}$ -

$$A^T = \begin{bmatrix} -2 & -2 \\ 4 & 4 \\ -2 & -2 \end{bmatrix}$$

Solving for $A^T y = 0$

$$\begin{bmatrix} -2 & -2 \\ 4 & 4 \\ -2 & -2 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0$$

$$\therefore -2y_1 - 2y_2 = 0 \quad 0 = xA$$

$$\therefore -2y_1 = 2y_2$$

$$\therefore y_1 = -y_2$$

$$\therefore y = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \Rightarrow \text{Left null space basis}$$

(4) row space basis $\Rightarrow$

By performing gaussian elimination on the matrix we can make any one row as zero's

$$\therefore \text{row space basis} = \{ (-2 \ 4 \ -2) \}$$

(5) Cross product of null spaces of A

$$v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

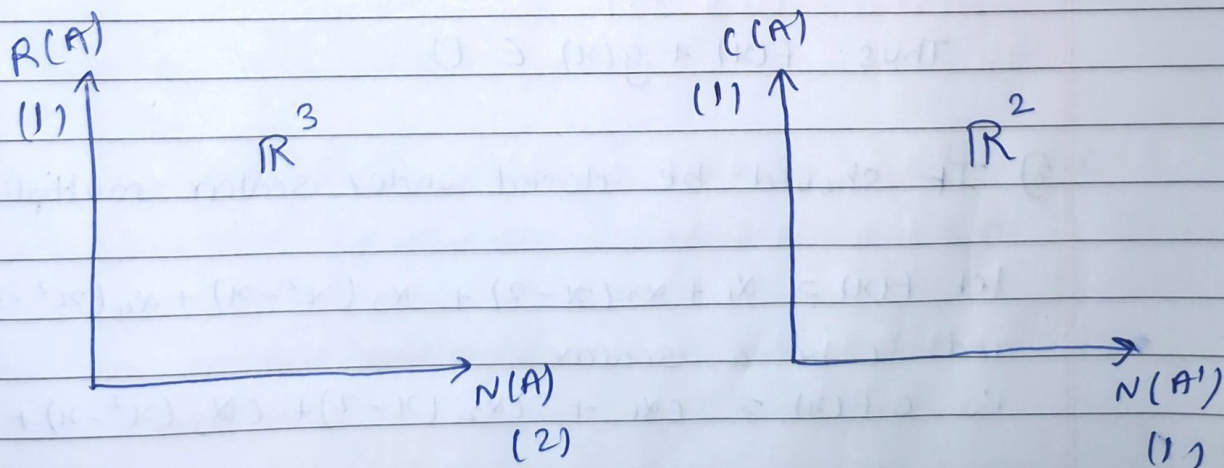
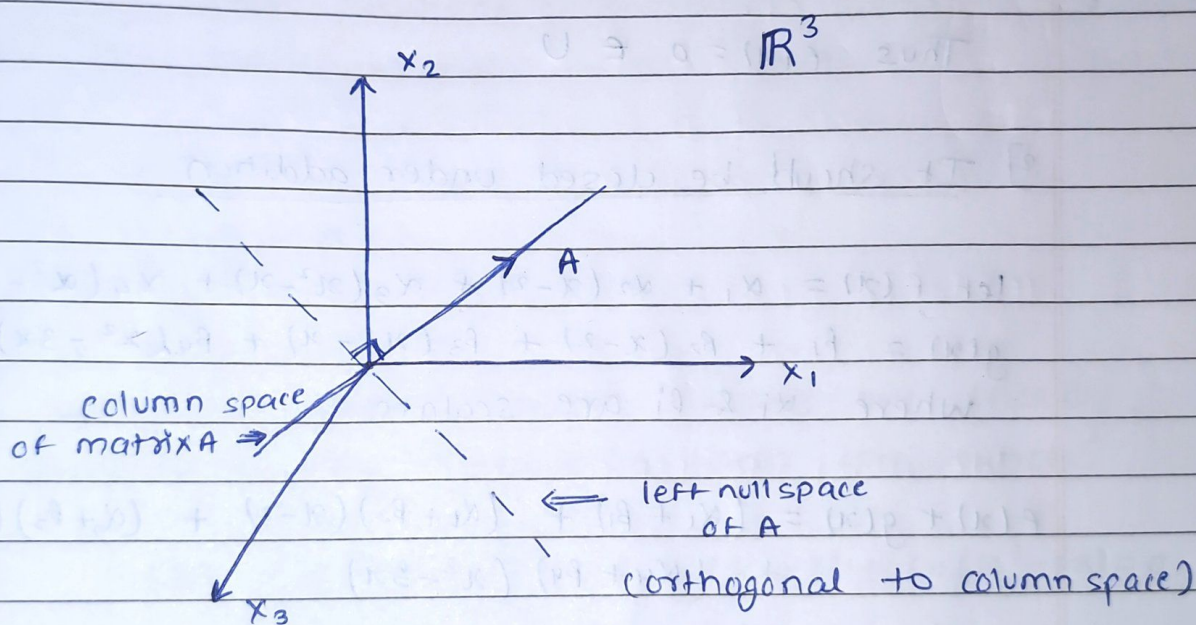
$$\therefore v_1 \times v_2 = \begin{vmatrix} i & j & k \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

$$= i(1-0) - j(2-0) + k(0+1)$$

$$= i - 2j + k$$

$$= \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

This vector is a scalar multiple of A



Q3]

(a) To show that U is a subspace of the vector space of functions $f(-\infty, \infty)$

We need to verify the following conditions:-

1.] Zero vector exists

The zero function $f(x) = 0$, which can be written as
 $f(x) = 0 + 0(x-2) + 0(x^2-x) + 0(x^2-3x)$

Thus $f(x) = 0 \in U$

2.] It should be closed under addition

Let $f(x) = \alpha_1 + \alpha_2(x-2) + \alpha_3(x^2-x) + \alpha_4(x^2-3x)$ and

$g(x) = \beta_1 + \beta_2(x-2) + \beta_3(x^2-x) + \beta_4(x^2-3x)$

where α_i & β_i are scalars.

$$f(x) + g(x) = (\alpha_1 + \beta_1) + (\alpha_2 + \beta_2)(x-2) + (\alpha_3 + \beta_3)(x^2-x) + (\alpha_4 + \beta_4)(x^2-3x)$$

Thus $f(x) + g(x) \in U$

3.] It should be closed under scalar multiplication

Let $f(x) = \alpha_1 + \alpha_2(x-2) + \alpha_3(x^2-x) + \alpha_4(x^2-3x)$

and c be a scalar.

$$\therefore c \cdot f(x) = c\alpha_1 + c\alpha_2(x-2) + c\alpha_3(x^2-x) + c\alpha_4(x^2-3x)$$

Thus $c \cdot f(x) \in U$

$\therefore U$ is a subspace of the vector space of functions $f(-\infty, \infty)$

(b) To obtain the basis of U and finding dimensions of U .

$$f(x) = \alpha_1 + \alpha_2(x-2) + \alpha_3(x^2-x) + \alpha_4(x^2-3x)$$

We can express this as a linear combination of functions:-

$$f(x) = \alpha_1(1) + \alpha_2(x-2) + \alpha_3(x^2-x) + \alpha_4(x^2-3x)$$

$\therefore$ The candidate basis functions are:-

(1) $f_1(x) = 1$

(2) $f_2(x) = x-2$

(3) $f_3(x) = x^2-x$

(4) $f_4(x) = x^2-3x$

We can check if these functions are linearly independent by forming a linear combination

$$c_1(1) + c_2(x-2) + c_3(x^2-x) + c_4(x^2-3x) = 0$$

Solving this shows

$$c_1 + (c_2 - c_3 - 3c_4)x + (c_3 + c_4)x^2 - 2c_2 = 0$$

$\therefore$ As zero function has all coefficients = 0

x^2 term $c_3 + c_4 = 0$

x term $c_2 - c_3 - 3c_4 = 0$

constant term $c_1 - 2c_2 = 0$

$\therefore$ solving this we get,

$$c_1 = -2c_4$$

$$c_2 = -c_4$$

$$c_3 = -c_4$$

$$c_4 = c_4$$

for all these equations, the only way to satisfy this system of equations is by

$$c_1 = c_2 = c_3 = c_4 = 0$$

confirming that these functions are linearly independent.

Thus, basis of U is $\{1, x-2, x^2-x, x^2-3x\}$, and the dimension of U is 4.

(c) Write $g(x) = 3x$ as a linear combination of your chosen basis vectors.

We want to express $g(x) = 3x$ as a linear combination of basis vectors.

$$g(x) = c_1(1) + c_2(x-2) + c_3(x^2-x) + c_4(x^2-3x)$$

This gives,

$$3x = c_1 + c_2(x-2) + c_3(x^2-x) + c_4(x^2-3x)$$

$$\therefore 3x = c_1 + c_2x - 2c_2 + c_3x^2 - c_3x + c_4x^2 - 3c_4x$$

Equating the coefficients of like powers of x , we get:

- coefficients of x^2 : $c_3 + c_4 = 0$
- coefficients of x : $c_2 - c_3 - 3c_4 = 3$
- coefficients of constant term :- $c_1 - 2c_2 = 0$

Solving these equations:-

$$1. \quad c_1 = 2c_2$$

$$2. \quad c_3 + c_4 = 0 \Rightarrow c_3 = -c_4$$

$$3. \quad c_2 - c_3 - 3c_4 = 3$$

$$\therefore c_2 + c_4 - 3c_4 = 3 \Rightarrow \text{(substituting from (2))}$$

$$\therefore c_2 - 2c_4 = 3$$

∴

Now, let's choose convenient values for the scalars.

Suppose, $c_4 = 0$ then $c_3 = 0$, and $c_2 = 3$. From $c_1 = 2c_2$ we get, $c_1 = 6$

Thus $g(x) = 3x$ can be written as:-

$$g(x) = 6(1) + 3(x-2)$$